

Benchmarking Project: Evaluating Variant Calling Performance at Different Depths of Coverage

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Important links

GitHub repo: https://github.com/sarah323/Benchmarking_project/tree/main

Google docs:

https://docs.google.com/document/d/1Mld0NdqcokXd8EURMM7633JPrF1_YLxEywvf_b3_xTLY/edit?tab=t.0#heading=h.dr77u7funubi

Supporting Files (Submitted Separately)

- hap.py benchmarking output files (for pipeline A and pipeline B)
- Full Pipeline A and B bash scripts and command logs
- Precision, Recall and F1 score plots for pipeline A and B
- Progress sheet report documenting all our meetings' agendas and progress
- Troubleshooting report
- NB: All these data are also available through our GitHub repo

1. Methods Summary

Dataset and Reference

Whole-exome sequencing (WES) data from the Genome in a Bottle (GIAB) HG002 sample were used for benchmarking. The original dataset was provided as a GRCh37 pre-aligned, duplicate-marked BAM generated using the Agilent SureSelect Human All Exon V5 capture kit. Variant benchmarking was restricted to chromosome 22. The GRCh38 reference genome and the GIAB v4.2.1 high-confidence truth set for HG002 were used throughout the analysis.

Two analytical pipelines (Pipeline A and Pipeline B) were implemented to evaluate the impact of alignment strategy and preprocessing decisions on variant detection accuracy across varying sequencing depths.

Pipeline A

Preprocessing and Alignment Strategy for pipeline A

The original BAM file was converted back to paired-end FASTQ files after name sorting. Reads were then realigned to the full GRCh38 reference genome using BWA-MEM, followed by coordinate sorting and duplicate marking with GATK MarkDuplicates. After duplicate marking, reads were restricted to chromosome 22 and further filtered to on-target regions defined by the SureSelect Human All Exon V5 BED file. The mean on-target depth prior to subsampling was approximately 221x.

Coverage Subsampling for pipeline A

To evaluate the effect of sequencing depth on variant detection, the on-target BAM was subsampled to four target coverages: 2x, 10x, 40x, and 80x. Subsampling was performed using samtools with a fixed random seed to ensure reproducibility. Mean depth was validated for each subset using samtools depth.

Variant Calling for pipeline A

Variants were called independently for each coverage level using GATK HaplotypeCaller, restricted to chromosome 22. The resulting VCF files were indexed and prepared for benchmarking. [Benchmarking](#)

Variant calling performance was evaluated using hap.py against the GIAB HG002 truth set, restricted to high-confidence regions on chromosome 22. Performance metrics included recall, precision, and F1 score, reported separately for SNPs and INDELS.

Pipeline B

Pipeline B aligned FASTQ reads directly to a chromosome 22-only reference using BWA-MEM, followed by duplicate marking with Picard. On-target regions were selected using the same SureSelect V5 BED file. Subsampling to identical target coverages (2x, 10x, 40x, 80x) was performed prior to variant calling using GATK HaplotypeCaller. Then hap.py was executed.

2. Benchmarking Results

1. Pipeline A

- SNP Performance Across Coverage Levels

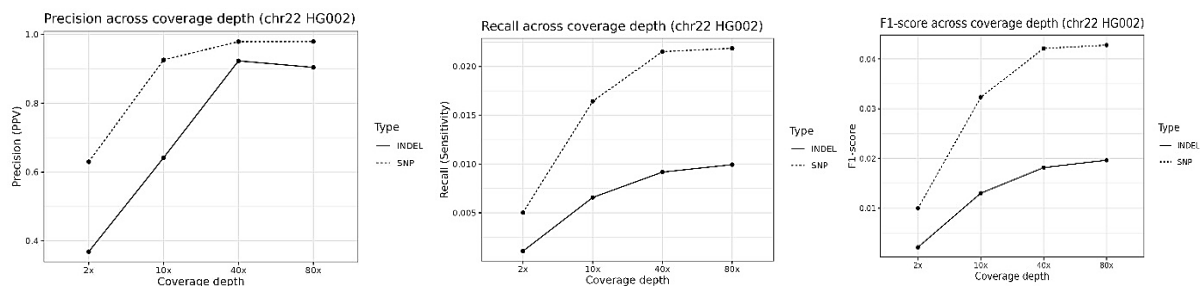
Coverage	Recall	Precision	F1 Score
2×	0.0050	0.6300	0.0100
10×	0.0164	0.9259	0.0323
40×	0.0215	0.9790	0.0421
80×	0.0219	0.9793	0.0428

SNP recall increased with coverage, particularly between 2x, 10x, and 40x then plateaued. Precision also improved at higher depths of 40x, then almost plateaued at 80X. Improvements in F1 score plateaued beyond 40× coverage as well.

- INDEL Performance Across Coverage Levels

Coverage	Recall	Precision	F1 Score
2×	0.0011	0.3684	0.0021
10×	0.0066	0.6418	0.0130
40×	0.0092	0.9231	0.0182
80×	0.0099	0.9041	0.0197

INDEL detection was substantially more challenging than SNP detection at all coverage levels. While precision improved markedly with depth, recall remained low, resulting in modest F1 scores even at high coverage.



2. Pipeline B

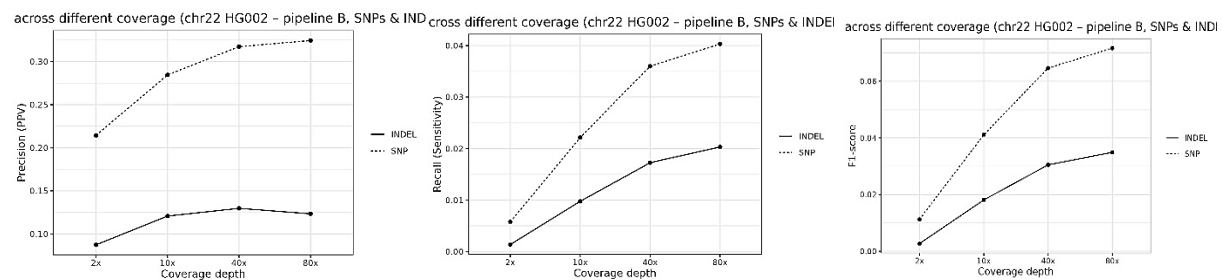
- SNP Performance Across Coverage Levels

Coverage	Recall	Precision	F1 Score
2x	0.0058	0.2144	0.0113
10x	0.0222	0.2846	0.0411
40x	0.0360	0.3173	0.0647
80x	0.0403	0.3244	0.0717

- INDEL Performance Across Coverage Levels

Coverage	Recall	Precision	F1 Score
2x	0.0014	0.0874	0.0027
10x	0.0098	0.1208	0.0181
40x	0.0173	0.1297	0.0305
80x	0.0203	0.1234	0.0349

For pipeline B, an improvement in recall and F1 score occurred with increasing coverages. Precision also improved for increasing coverage, but it was well noted that precision results for pipeline B are significantly lower than pipeline A.



Pipeline A and Pipeline B produce different benchmarking results primarily because of how read alignment is performed. In Pipeline A, reads are first aligned to the entire GRCh38 genome, allowing ambiguous or repetitive reads to be correctly placed, soft-clipped, or assigned low mapping quality based on full-genome context. Only after this realistic alignment step is the data restricted to chromosome 22, meaning that downstream variant calling reflects true genome-wide mapping uncertainty and more accurately represents real sequencing conditions.

In contrast, Pipeline B aligns reads directly to chromosome 22 only, removing the competitive context of the rest of the genome. As a result, reads that would normally map to other chromosomes are forced onto chr22, leading to artificially high alignment rates and an increase in false-positive variant calls. This misplacement distorts precision, recall, and F1 scores, making Pipeline B less reliable for benchmarking and explaining the observed discrepancies between the two pipelines.

3. Interpretation and Discussion (for pipeline A mainly)

Effect of Coverage Depth on Variant Detection Accuracy

Increasing sequencing depth led to clear improvements in recall and F1 score for both SNPs and INDELs, particularly when moving from very low coverage (2×) to moderate coverage (10×–40×). Precision improved rapidly and remained high at higher depths, especially for SNPs. However, gains in recall were modest beyond 40× coverage, indicating diminishing returns.

SNPs vs INDELs

Across all depths, SNPs were detected with substantially higher accuracy than INDELs. INDEL calling showed persistently low recall despite high precision at higher coverage, reflecting the inherent difficulty of accurately detecting insertion and deletion events using short-read sequencing and standard variant calling approaches.

Trade-off Between Sequencing Cost and Accuracy

The results demonstrate that increasing coverage substantially improves variant detection accuracy up to a point, but the benefit per additional sequencing depth decreases beyond moderate coverage. While 80× coverage marginally improved performance compared to 40×, the improvement in F1 score was minimal. From a cost–benefit perspective, 40× coverage appears to provide a reasonable balance between sequencing cost and variant calling accuracy for SNP detection in WES data.

Does Higher Depth Always Improve Performance?

Higher depth does not always lead to proportionally better performance. Although precision remains high, recall plateaus at higher depths, particularly for SNPs, and remains limited for INDELs. These findings suggest that factors such as read alignment quality, capture efficiency, and variant caller limitations play a significant role in performance, beyond sequencing depth alone.

4. Conclusion

This benchmarking analysis demonstrates that a full-reference alignment strategy (Pipeline A) yields highly precise variant calls on chromosome 22, with performance improving as sequencing depth increases. However, diminishing returns are observed beyond 40× coverage, particularly for SNPs, and INDEL detection remains challenging even at high depth. These results highlight the importance of balanced experimental design and support the use of moderate coverage levels for cost-effective and accurate WES variant detection.